

Assignment - 1 Solutions

1. Convert $(1110.011)_2$ to Decimal

Integer Part:

$$1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 8 + 4 + 2 + 0 = 14$$

Fractional Part:

$$0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 0 + 0.25 + 0.125 = 0.375$$

Result:

$$14 + 0.375 = 14.375$$

2. Convert $(72945)_{10}$ to binary, BCD and Excess-5

Binary - $(10001110011110001)_2$

BCD - $(0111\ 0010\ 1001\ 0100\ 0101)_{\text{BCD}}$

Ex-5 - $(1100\ 0111\ 1110\ 1001\ 1010)_{\text{Ex-5}}$

Binary not equal to BCD [Proved]

3. Convert $(123.8125)_{10}$ to Base-6

Integer Part:

$$123 \div 6 \text{ repeatedly (remainders)} \rightarrow 323$$

Fractional Part:

$$0.8125 \times 6 = 4.875 \rightarrow 4$$

$$0.875 \times 6 = 5.25 \rightarrow 5$$

Result: 323.45

4. Convert $(2F8.BE)_{16}$ to Octal

1. Hex to Binary: $2F8.BE \rightarrow 0010\ 1111\ 1000\ .\ 1011\ 1110$

2. Group into 3-bit chunks: $001\ 011\ 111\ 000\ .\ 101\ 111\ 100 \rightarrow 1\ 3\ 7\ 0\ .\ 5\ 7\ 4$

Result: 1370.574_8

5. Multiply $(132)_5 \times (24)_5$ in Base 5

$$\begin{array}{r} 132 \\ \times 24 \\ \hline 1133 \\ 3140 \\ \hline 4323 \end{array}$$

6. Divide $(56214306)_7$ by $(620)_7$

- Quotient: 00064004
- Remainder: 566

7. Subtract -45 from 23 in 8-bit 2's Complement

$$23 - (-45) = 23 + 45$$

1. 2's Complement of 45: 00101101
2. Add to 23: $00010111 + 00101101 = 0100\ 0100 = 68$
3. Result: (No overflow)

8. Add -89 and 74 in 12-bit 1's Complement

1. 1's Complement of -89: 111110100110
2. Add to 74: $111110100110 + 000001001010 = 11111110000$
3. Final Result: (No overflow)

9. Total Cost and Remaining Money

- Laptop: $10110101_2 = 181$
- Monitor: $B2_{16} = 178$
- Keyboard: $56_8 = 46$
Total Cost: $181 + 178 + 46 = 405$
Your Money: $100100110_2 = 294$
Deficit: $405 - 294 = 111$ dollars

10. Party Expenses

- Cake: $1101_2 = 13$
- Balloons: $24_8 = 20$
- Decorations: $1A_{16} = 26$
- Snacks: $5 \times 101_2 = 25$
Total Cost: $13 + 20 + 26 + 25 = 84$
Your Money: $100011_2 = 35$
Deficit: $84 - 35 = 49$ dollars